

CSE - 641 Computer Vision

Weekly Project Report – 5



**Ahmedabad
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Genotype-to-Phenotype Modeling of Alzheimer's Disease Using MRI Radiomics and Polygenic Risk Scores

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Abstract—Alzheimer disease is a disorder that affects the brain leading to loss of memory and progressive impairment of thinking ability. Studies indicate that genetic factors as well as changes in brain structure are significant in the pathogenesis of Alzheimer disease. However, the correlation between inheritable genetic risk and structural brain change is not fully understood.

This project examines the association between genetic vulnerability and brain structure using Polygenic Risk Scores (PRS) and MRI-based radiomic features. Important brain regions including the hippocampus and entorhinal cortex are analyzed using MRI scans from the Alzheimer Disease Neuroimaging Initiative (ADNI) dataset.

PRS values are computed to estimate inherited genetic risk, and machine learning models are used to identify relationships between genetic risk and MRI features. The goal is to better understand the role of genetic factors in early brain changes related to Alzheimer disease, potentially before clinical symptoms appear.

I. INTRODUCTION

Alzheimer disease is an advanced disorder of the brain which gradually impairs the memory, ability to think and general brain performance. It is connected with when brain tissue is gradually lost throughout time. These changes are usually studied by structural magnetic resonance imaging (MRI), particularly in those parts of the brain that become affected at an early stage in the disease, the hippocampus, entorhinal cortex and temporal lobes [1]. Meanwhile, genome wide association studies of large sizes have enhanced the insights into the genetic etiology of Alzheimer. The genetic risk variants that have been identified through such studies are too many and predispose an individual with the probable cause of the disease [2]. These variants may be used together in a polygenic risk score (PRS) i.e., to sum up inherited genetic risk in a single value [3]. Both MRI markers and PRS can be used to understand the disease, but this is in two dimensions- brain structure and genetic risk. There have been many research studies that attempted to improve early detection of Alzheimer disease with use of brain imaging methods. Structural magnetic resonance imaging is one of the methods that have been used majorly as it is capable of showing the changes that take place in the brain regions which are affected by disease. One of the affected regions is the hippocampus which is part of the brain that is affected in the Alzheimer disease as it is the part associated with memory. Previous studies were interested in the size of the brain regions. However with the use of the radiomics method, more quantitative features of the MRI images may be obtained that may potentially reveal more details. The authors of the paper [4] would like to create a model of radiomics using MRI images of the hippocampus of the brain to aid in the identification of patients suffering from Alzheimer disease. This would be achieved by the use of data from various cohorts to ensure the reliability of the model. Radiomic features would be achieved by the use of MRI images of the hippocampus. This would then be used to create a model that would aid in the differentiation of patients suffering from Alzheimer disease and healthy patients. This would be achieved by the use of MRI images and the expression of genes present in the blood.

This would then attempt to establish a connection between the genetic features and the biological processes aiming to have a better understanding of the changes taking place in the brain as shown by MRI images.

Recent studies have explored radiomics-based analysis of MRI scans to extract quantitative features that may reveal subtle changes not visible through conventional measurements. For example, Xia et al. [4] developed a hippocampal radiomics model using multi-cohort MRI data to differentiate Alzheimer patients from healthy controls and to investigate biological mechanisms linked to imaging features.

Combining radiomic features with genetic information may help improve early detection and understanding of disease progression. The summary of the selected parameters from the ADNI database is given in the Table 1.

II. DATASET DESCRIPTION

The data used in this project is obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI), a multi-centered longitudinal large-scale study established to identify and validate biomarkers for the early detection and monitoring of Alzheimer's disease progression. The study participants are categorized into these primary groups:

- Cognitively Normal (CN)
- Significant Memory Concern (SMC)
- Early Mild Cognitive Impairment (EMCI)
- Late Mild Cognitive Impairment (LMCI)
- Alzheimer's Disease (AD)

ADNI provides multimodal data, including:

- Structural Magnetic Resonance Imaging (MRI)
- Positron Emission Tomography (PET) scans (FDGPET and Amyloid PET)
- Cerebrospinal Fluid (CSF) biomarker values
- Genetic data (e.g., APOE genotype)
- Neuropsychological assessment scores (e.g., MMSE, CDR, ADAS-Cog)

The data were collected across multiple phases based on the years it was collected, namely ADNI-1, ADNI-GO, ADNI-2, ADNI-3, ADNI 4.

These phases enable longitudinal analysis of disease progression over time. The dataset is cross-standardized across acquisition sites, ensuring consistency and reliability. Due to its comprehensive and longitudinal nature, the ADNI dataset is widely used in machine learning and deep learning research for classification, progression prediction, and biomarker discovery in Alzheimer's disease studies.

III. LITERATURE REVIEW

A. Longitudinal structural MRI-based deep learning and radiomics features for predicting Alzheimer's disease progression

The authors in this paper show how a longitudinal MRI-based paradigm can be utilized to forecast the subjects who will progress to Alzheimer disease from mild cognitive impairment (MCI). They build a twostage deep learning model on the

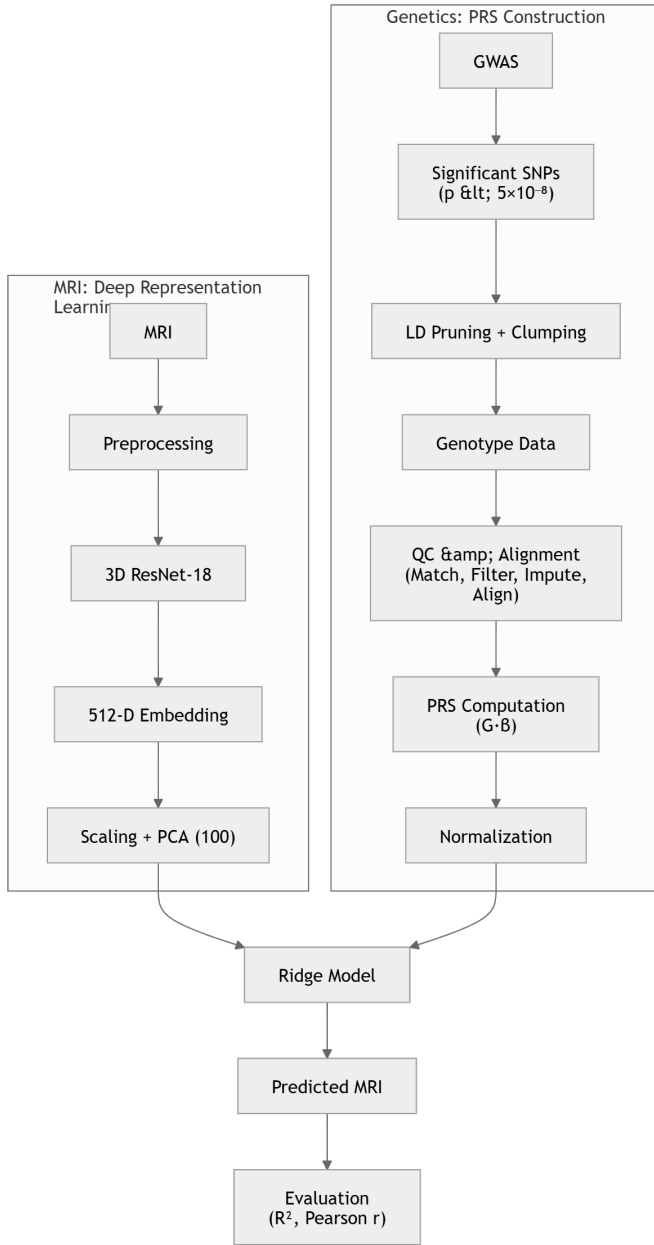


Fig. 1: Flowchart of the performed end-to-end pipeline.

T1-weighted images of the Alzheimer Disease Neuroimaging Initiative dataset. They first train a 3D ResNet18 model on individual visit MRI volumes to enable it to learn spatial brain features related to disease risk with a survival-based ranking loss. The learned feature representations are then used as input to a time-varying LSTM that trains to learn multiple MRI scans over time, accounting for irregular time intervals between visits. The attention mechanism identifies the most relevant time points. They also discard gray matter radiomics features like shape, intensity, and texture features, and discard them using PCA, and then combine them with deep features. The final model is being placed within a survival analysis framework to estimate individualized risk of time-to-

conversion, rather than just binary classification.

B. Predicting Conversion from Mild Cognitive Impairment to Alzheimer's Disease Using a Vision Transformer and Hippocampal MRI Slices

The deep learning models are employed to predict the development of Mild Cognitive Impairment (MCI) to Alzheimer's Disease (AD) from MRI scans. Most of the existing studies were conducted using Convolutional Neural Networks (CNNs) because they possess the capability to learn spatial information from images of the brain. Some studies have been demonstrated to possess high accuracy, but there were problems such as data leakage, where data from the same patient was used for training and testing, which could potentially lead to biased results. Recently, Vision Transformers (ViTs) have been proposed for image processing. Although Vision Transformers have been demonstrated to possess high accuracy for Alzheimer's classification, few studies have been conducted to explore their application for predicting MCI to AD conversion. This paper continues from the existing studies by employing a Vision Transformer model that was pretrained on hippocampal MRI slices. It strictly separates data for each subject and compares the results to a 3D CNN model. The results indicate that Vision Transformers possess the capability to produce similar results to CNN models. In [5], genetic markers as single nucleotide polymorphisms (SNP) are compared to brain imaging measurements carried out by the use of MRI scans to perform this analysis. A statistical test is then used to find significant correlations between genetic variants and specific neuroimaging features regarding the Alzheimer disease. Polygenic Risk Scores provide an understanding of how combination of genes of a person may influence chance of developing Alzheimer's disease. The authors of the [6] explored many case control and by proxy studies data through a genome wide association method to determine genetic differences associated with Alzheimer's disease. The research shows that when PRS used in combination with APOE carrier status could stratify individuals on the risk and the median age at onset of high-risk individuals differs. This paper brings attention to the fact that it is better to combine the numerous genetic risk factors into one score that would further predict the disease susceptibility. Though the study mainly concentrates on the genetic side it also forms the basis of the association of genetic risk with alteration of the brain which show that early change in the structure of the brain could occur in people with high PRS. In the context of our project this provides the reasoning behind the integration of PRS and MRIbased radiomic features in the effort to investigate the development of cumulative genetic risk into subtle phenotypes in hippocampal and entorhinal cortex structures which would possibly support the earlier detection of the advancement of Alzheimer disease.

IV. METHODOLOGY

The goal of this project is to examine the relationship between genetic risks and brain structure by integrating Polygenic Risk Scores with features derived from MRI scans. First,

TABLE I: Summary of Selected ADNI Dataset Parameters

Parameter	Selection
ADNI Phases	ADNI-2
Imaging Modality	Structural MRI
MRI Sequence	T1-weighted MPRAGE
Field Strength	3 Tesla (3T)
Visit Type	Baseline and Screening
Subject Groups	CN, EMCI, LMCI, AD
Image Format	NIfTI (.nii)
Genetic Data	APOE Genotype

datasets were collected that contained both genetic information and corresponding MRI scans. The individuals that were common to both datasets were identified using a common identifier, referred to as IID. These individuals were then used for analysis.

The next steps were to preprocess the genetic information to obtain PRS values, which represent the overall risk based on various genetic weights. The MRI scans were then preprocessed by extracting features, whereby each scan is converted into a fixed-length numerical vector that represents brain features.

Finally, a new dataset is formed by combining the preprocessed PRS values with MRI features based on subject ID. The overall goal is to create a predictive model that predicts MRI features using PRS values as input. The overall strategy is to use 5-fold cross-validation.

For each fold, MRI features are standardized using PCA to reduce dimensionality. A Ridge Regression model is then used to create a predictive model using PRS values as input features and MRI features as output features.

V. WORK COMPLETED IN WEEK-5

We have employed the FastSurfer framework to achieve deep learning-based brain segmentation on NIfTI format structural MRI scans. For each subject, we produced accurately computed and detailed anatomical labels; specifically targeting the extraction of the hippocampus and entorhinal cortex, as both anatomical regions will show clear signs of atrophy with Alzheimer’s disease and are also typically two of the earliest regions of the brain to be impacted by disease. The additional output of clearly defined binary labels was created by isolating the hippocampus and entorhinal cortex regions using the pre-defined label indices associated with these two structures. Each binary mask created was stored in numpy format for feature extraction and subsequent downstream processing. Focusing on these two clinically relevant brain regions ensures that our pipeline identifies clinically relevant biomarkers that will lead to enhancements in the performance and interpretability of the ML/DL models for early prediction of the Alzheimer’s disease process.

VI. PLAN FOR WEEK-6

For the upcoming Week 6, we plan to conduct regression analysis on the segmented MRI data and also improve the modeling pipeline as follows:

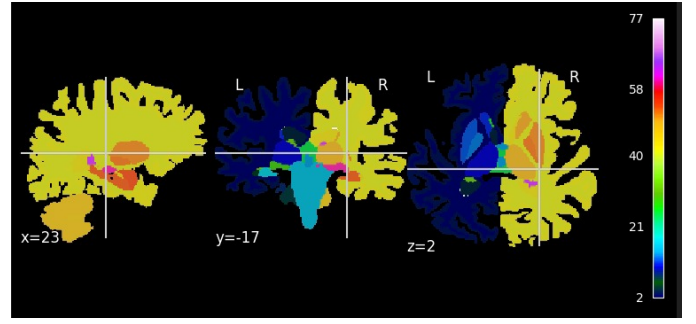


Fig. 2: Segmented 3D brain MRI highlighting different anatomical regions across multiple views

- Conduct regression analysis on the segmented brain region data rather than the global features of the MRI data.
- Investigate region-specific relationships between PRS and key brain areas such as the hippocampus and entorhinal cortex.
- Investigate relevant research papers to improve and optimize the generative pipeline.
- Experiment with advanced modeling techniques to improve the modeling of non-linear relationships between genetic and imaging data.
- Improve the preprocessing and feature selection strategy for better model performance.

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